THE OPEN UNIVERSITY OF SRI LANKA

DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

BACHELOR OF SOFTWARE ENGINEERING

# EEI6373 – Performance Modelling

# Mini Project Report

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# 1. System Description and Performance Goals

## 1.1 System Description

The system studied is an outpatient department (OPD) healthcare delivery process. The model represents a patient moving through two dependent service stages: registration and doctor consultation.

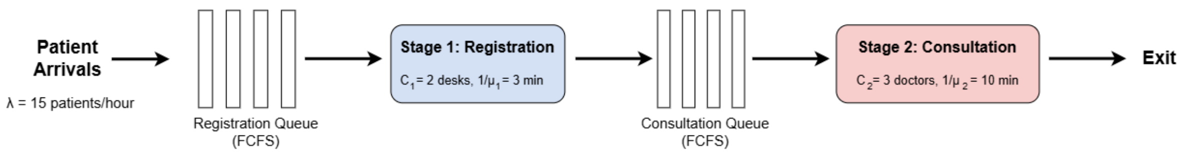


*Figure 1: Two-stage tandem queueing model of the OPD system.*

At registration, a patient waits if both desks are busy. After registration, the patient joins the consultation queue and waits for an available doctor. The two stages are connected, so delay at one stage can affect the next stage. Multiple parallel servers also create resource-allocation effects. These characteristics make the OPD suitable for studying waiting time, utilisation, congestion and capacity limits.

The model is a hypothetical OPD rather than a model of a named hospital. The dataset is synthetic and the parameter values are modelling assumptions, not measurements from a real hospital.

## 1.2 Performance Goals

The primary objective is to reduce patient waiting time. Secondary objectives are to identify the bottleneck, understand how waiting changes as workload increases, and determine whether adding doctors provides worthwhile improvement.

| Objective | Metric | Interpretation |
|---|---|---|
| Minimise patient waiting | Wq,total – average total waiting time | Lower is better |
| Reduce registration delay | Wq,reg | Lower is better |
| Reduce doctor delay | Wq,doc | Lower is better |
| Improve end-to-end response | W – average time in system | Lower is better |
| Identify bottleneck | ρreg and ρdoc | Higher utilisation indicates greater load |
| Evaluate capacity | Wq,total versus λ and c2 | Compare configurations |

An average total waiting time below 15 minutes is used as a project design target. This is an explicit modelling assumption, not a claim about an official hospital standard.

# 2. Modelling Approach and Assumptions

## 2.1 Modelling Technique

Discrete event simulation (DES) is the main modelling technique. The OPD is modelled as a two-stage tandem queueing system. The registration stage is modelled as an M/M/2 queue, while the consultation stage is modelled as an M/M/3 queue with three parallel doctors. DES is suitable because the project studies individual patient waiting times and allows controlled what if experiments on arrival rate and doctor capacity.

| Symbol | Parameter | Base value |
|---|---|---|
| $\lambda$ | Patient arrival rate | 15 patients/hour (mean inter-arrival 4 min) |
| $c_1$ | Registration desks | 2 |
| $1/\mu_1$ | Mean registration service time | 3 min |
| $c_2$ | Doctors | 3 |
| $1/\mu_2$ | Mean consultation service time | 10 min |
| N | Patients per run | 500 |

## 2.2 Simulation Procedure

Inter-arrival times and service times are generated from exponential distributions. A minimum-heap stores the next available time of each server. A patient starts service at the later of the patient's arrival/transfer time and the earliest server-free time. After registration, patients enter the doctor stage in registration-completion order. The program records waiting time, service time and total time in the system for every patient.

A fixed random seed of 42 makes the results reproducible. The same seed is used for scenario comparisons, so differences are primarily caused by the parameter being changed.

## 2.3 Assumptions

- **A1 –** Arrivals follow a Poisson process, giving exponential inter-arrival times.
- **A2 –** Registration and consultation service times are exponentially distributed.

- **A3 –** Patients are served first-come-first-served at both stages.
- **A4 –** Servers are identical and continuously available.
- **A5 –** Queue capacity is sufficient; patients do not leave because of long waits.
- **A6 –** The arrival rate is stationary; time-of-day peaks are not modelled.
- **A7 –** The simulation starts empty and no warm-up period is removed.

These assumptions simplify the real OPD. Exponential service can create more variability than some real services, stationary arrivals ignore morning peaks, and an empty-start finite run can underestimate steady-state waiting.

## 2.4 Key Formulas and Sample Calculations

This subsection presents the main performance formulas used in the OPD model together with representative calculations for the base scenario.

**Arrival Rate**

The mean inter-arrival time is 4 minutes. Therefore,

- **$\lambda = 60 / 4 = 15$ patients per hour**

Thus, on average, 15 patients arrive per hour.

**Registration Service Rate and Utilisation**

The mean registration service time is 3 minutes. The service rate of one registration desk is,

- **$\mu_1 = 60 / 3 = 20$ patients per hour**

With two registration desks ($c_1 = 2$), the utilisation is:

- **$\rho reg = \lambda / (c_1 \times \mu_1)$**
- **$\rho reg = 15 / (2 \times 20) = 0.375 \approx 37.5\%$**

The simulation produced a registration utilisation of approximately **38.0%**, which is consistent with this analytical estimate.

**Doctor Service Rate and Utilisation**

The mean consultation time is 10 minutes. The service rate of one doctor is:

- **$\mu_2 = 60 / 10 = 6$ patients per hour**

With three doctors ($c_2 = 3$), the theoretical utilisation is:

- **$\rho doc = \lambda / (c_2 \times \mu_2)$**
- **$\rho doc = 15 / (3 \times 6) = 0.833 \approx 83.3\%$**

The simulation reported a doctor utilisation of approximately **72.9%**. The difference occurs because the simulation is a finite run of 500 patients, begins from an empty system, and includes random variability.

**Stability Check for the High-Load Scenario**

For the 20 patients/hour experiment:

$$\rho = 20 / (3 \times 6) = 1.11$$

Since $\rho > 1$, the consultation stage is theoretically unstable; therefore, the finite-run waiting-time result is not a steady-state performance measure.

**Sample Patient Calculation**

For one patient in the dataset:

- Arrival time = 12.00 min
- Registration start = 13.20 min
- Registration end = 15.10 min
- Consultation start = 20.00 min
- Consultation end = 31.50 min

Registration waiting time:

$$W_{reg} = 13.20 - 12.00 = 1.20 \text{ min}$$

Doctor waiting time:

$$W_{doc} = 20.00 - 15.10 = 4.90 \text{ min}$$

Total waiting time:

$$W_{total} = W_{reg} + W_{doc} = 1.20 + 4.90 = 6.10 \text{ min}$$

Total time in the system:

$$T_{system} = 31.50 - 12.00 = 19.50 \text{ min}$$

These calculations illustrate how the patient level metrics stored in **opd_dataset.csv** are derived from the recorded event times.

# 3. Data Description and Methodology

## 3.1 Data Acquisition

No operational hospital log was used. The project uses synthetically generated data, which is permitted by the brief. The base inputs 15 patients/hour, 3 minute registration and 10-minute consultation are illustrative modelling assumptions selected to create a meaningful but manageable OPD workload. They are not empirical estimates.

The base run produces opd_dataset.csv with one record per patient. The program also produces opd_summary.csv with one row per scenario and four report figures.

## 3.2 Dataset Structure

| Field | Meaning |
|---|---|
| Patient_ID | Patient sequence number |
| Arrival_Time | Patient arrival time |
| Reg_Wait | Waiting before registration |
| Registration_Start / End | Registration start and finish |
| Reg_Service_Time | Registration duration |
| Cons_Wait | Waiting before doctor |
| Consultation_Start / End | Consultation start and finish |
| Consultation_Service_Time | Consultation duration |
| Total_Waiting_Time | Reg_Wait + Cons_Wait |
| Time_in_System | Consultation_End − Arrival_Time |

## 3.3 Experimental Design

A one factor at a time design is used to isolate the effect of each control variable. The base scenario uses an arrival rate of 15 patients/hour, two registration desks, and three doctors. Experiment A varies the arrival rate while keeping doctor capacity fixed at three, whereas Experiment B varies the number of doctors while keeping the arrival rate fixed at 15 patients/hour.

| Experiment | Variable | Values |
|---|---|---|
| Base | None | $\lambda$=15/hr, c1=2, c2=3 |
| A – workload | Arrival rate $\lambda$ | 10.0, 12.0, 15.0, 17.1, 20.0/hr |
| B – capacity | Doctors c2 | 2, 3, 4, 5 |

# 4. Detailed Analysis and Findings

## 4.1 Base Scenario

| Metric | Registration | Consultation | Whole System |
| --- | --- | --- | --- |
| Average waiting time | 0.42 min | 6.52 min | 6.95 min |
| Utilisation | 38.0% | 72.9% | — |
| Average time in system | — | — | 19.54 min |

The doctor stage is the clear bottleneck. Doctor waiting is 6.52 minutes compared with only 0.42 minutes at registration, so approximately 94% of total waiting occurs before consultation. Doctor utilisation is also much higher than registration utilisation (72.9% versus 38.0%).

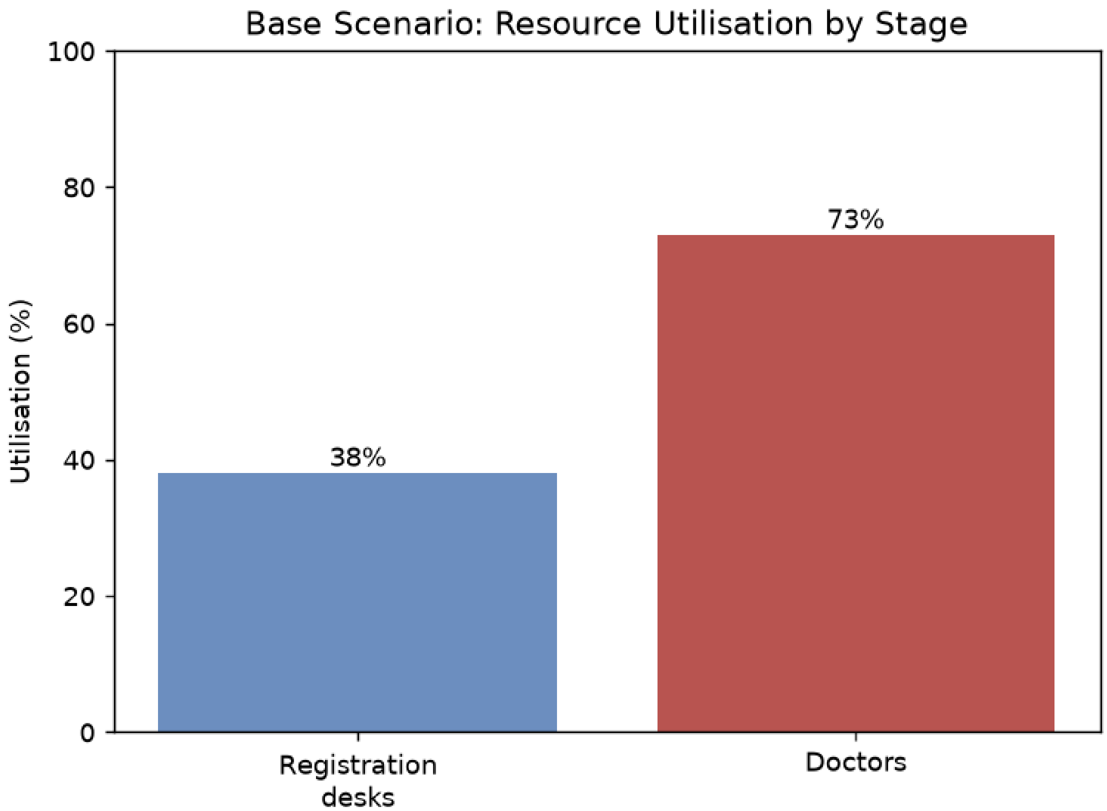


*Figure 2. Base-scenario resource utilisation.*

## 4.2 Experiment A – Increasing Arrival Rate

| **Arrival rate (patients/hr)** | **Patients** | **Avg. Reg. Wait (min)** | **Avg. Doctor Wait (min)** | **Avg. Total Wait (min)** | **Avg. Time in System (min)** | **Registration Utilisation** | **Doctor Utilisation** |
|---|---|---|---|---|---|---|---|
| 10.0 | 500 | 0.17 | 1.84 | 2.01 | 14.60 | 25.9% | 49.7% |
| 12.0 | 500 | 0.24 | 3.31 | 3.55 | 16.15 | 30.9% | 59.4% |
| 15.0 | 500 | 0.42 | 6.52 | 6.95 | 19.54 | 38.0% | 72.9% |
| 17.1 | 500 | 0.61 | 10.29 | 10.90 | 23.49 | 42.8% | 82.1% |
| 20.0 | 500 | 0.98 | 22.50 | 23.48 | 36.07 | 48.0% | 92.2% |

Waiting time increases non-linearly as workload rises. From 15 to 20 patients/hour, the arrival rate increases by about 33%, while average total waiting increases from 6.95 to 23.48 minutes. Average time in the system also increases from 19.54 to 36.07 minutes. The deterioration is concentrated at the doctor stage.

The 20 patient/hour result is not a steady-state result. With three doctors and a 10-minute mean consultation time, $\rho = 20/(3 \times 6) = 1.11$, so the offered workload exceeds capacity and no steady state exists. The finite 500-patient value should therefore be interpreted as a finite-run result rather than a steady-state performance measure. At 17.1 patients/hour, the theoretical doctor utilisation is $\rho = 17.1/(3 \times 6) = 0.95$, while the measured utilisation in the finite simulation is 82.1%. This makes 17.1 patients/hour a useful warning level because the theoretical workload is already close to capacity.

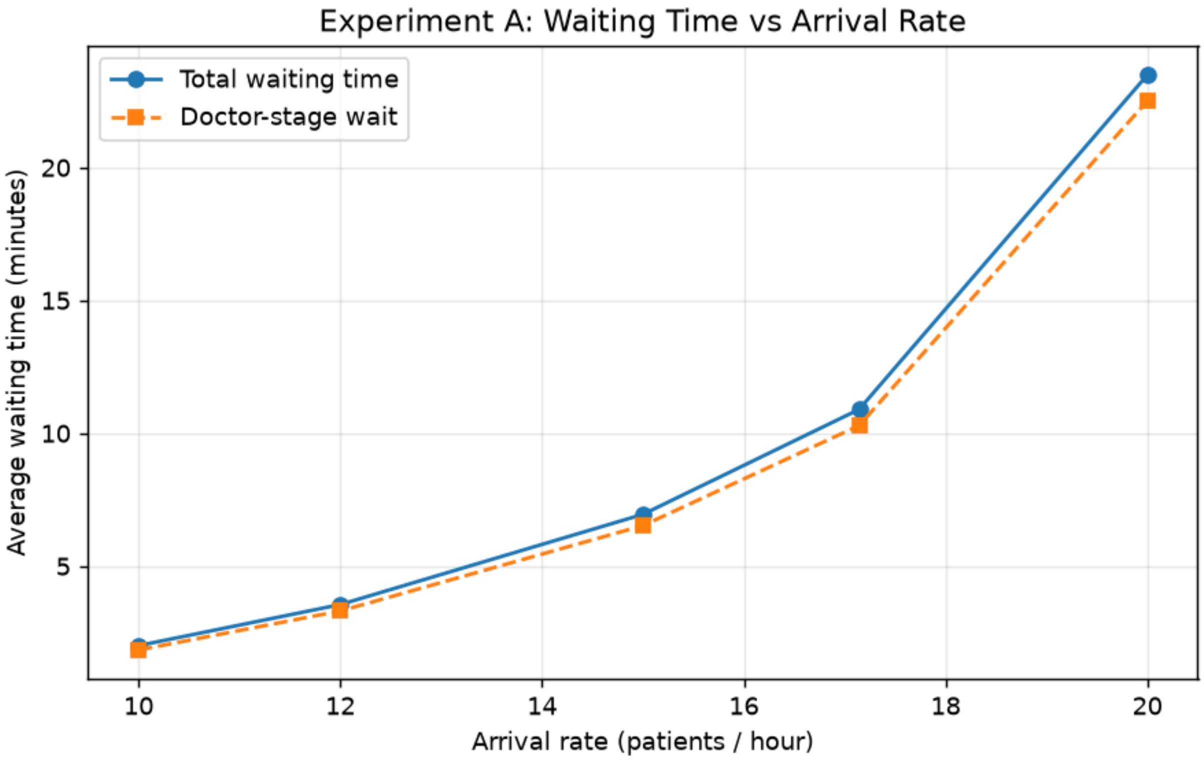


*Figure 3. Effect of arrival rate on average waiting time.*

## 4.3 Experiment B – Number of Doctors

| Doctors | Patients | Avg. Reg. Wait (min) | Avg. Doctor Wait (min) | Avg. Total Wait (min) | Avg. Time in System (min) | Registration Utilisation | Doctor Utilisation |
|---|---|---|---|---|---|---|---|
| 2 | 500 | 0.42 | 73.33 | 73.76* | 86.35 | 34.3% | 98.8% |
| 3 | 500 | 0.42 | 6.52 | 6.95 | 19.54 | 38.0% | 72.9% |
| 4 | 500 | 0.42 | 1.53 | 1.95 | 14.54 | 38.5% | 55.4% |
| 5 | 500 | 0.42 | 0.31 | 0.73 | 13.32 | 38.5% | 44.3% |

Adding doctors provides the largest benefit when the consultation stage is close to or beyond capacity. Increasing the number of doctors from two to three reduces average total waiting time by 66.81 minutes, from 73.76 to 6.95 minutes. Adding a fourth doctor provides a further reduction of 5.00 minutes, while adding a fifth doctor reduces waiting by only 1.22 minutes. This demonstrates diminishing returns from additional doctor capacity. The two-doctor configuration is theoretically unstable because $\rho = 15/(2 \times 6) = 1.25 > 1$, meaning that the offered workload exceeds the available consultation capacity.

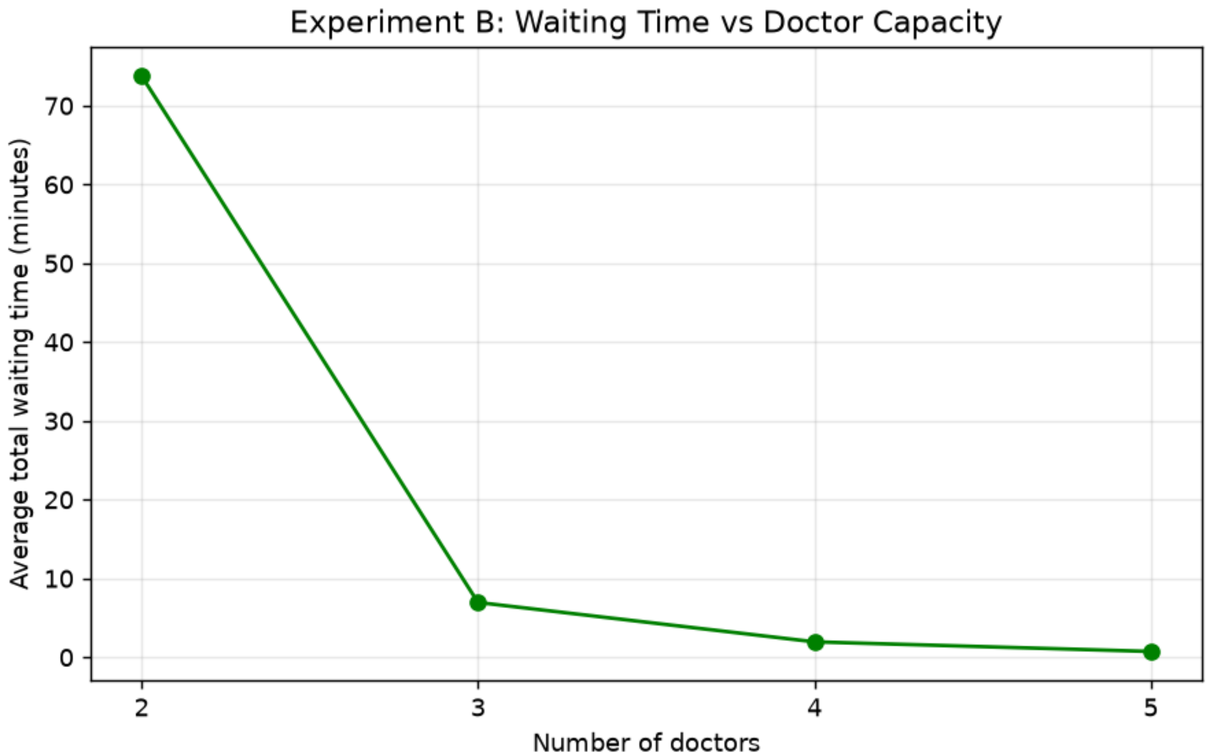


*Figure 4. Effect of doctor capacity on average total waiting time.*

## 4.4 Distribution of Patient Waiting Time

| **Statistic** | **Value** |
|---|---|
| Mean | 6.95 min |
| Median | 1.95 min |
| 90th percentile | 19.62 min |
| 95th percentile | 27.66 min |
| Maximum observed | 58.54 min |
| Patients waiting <5 min | 61.8% |
| Patients waiting >20 min | 9.6% |

The average does not fully describe patient experience. The median waiting time is 1.95 minutes, compared with a mean of 6.95 minutes, while the maximum observed wait is 58.54 minutes. The histogram shows a concentration of patients at low waiting times with a long right tail, indicating a strongly right-skewed distribution. This supports the use of percentile-based service targets in addition to average waiting time. Percentiles were calculated from opd_dataset.csv using NumPy's default linear interpolation method.

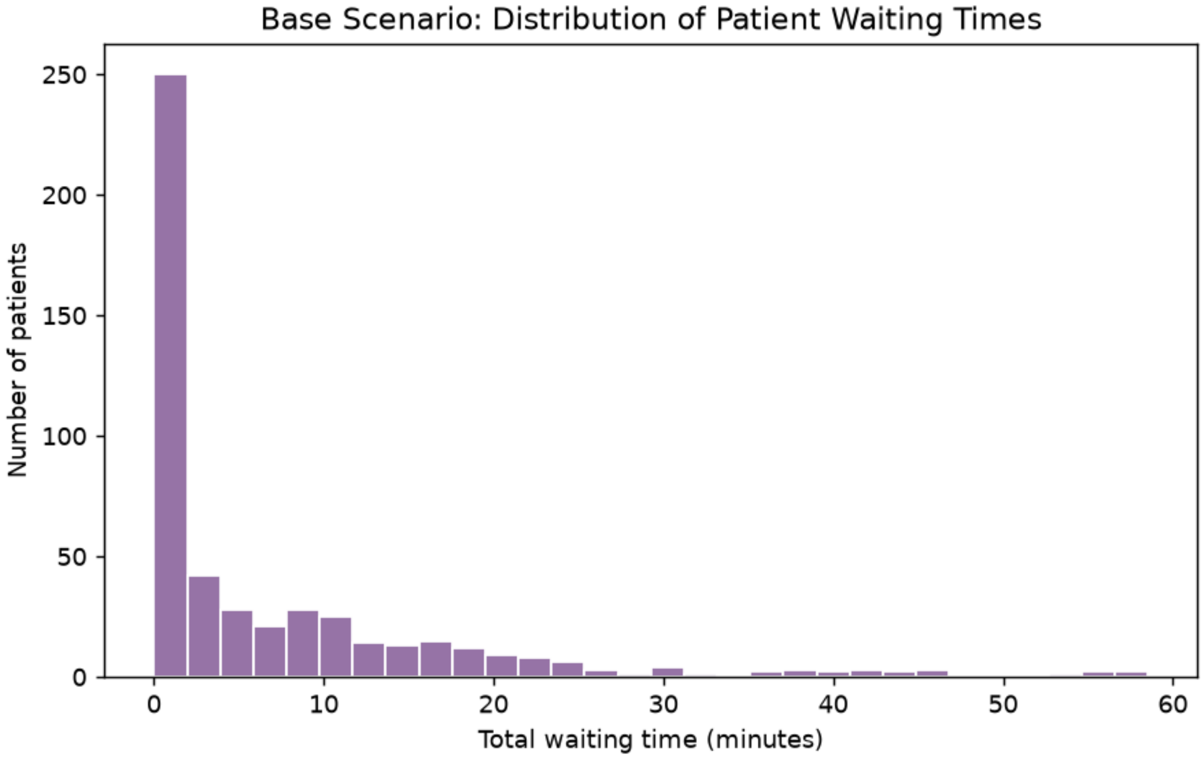


*Figure 5: Distribution of total patient waiting times in the base scenario (500 patients).*

## 5. Performance Tuning and Recommendations

The results support targeted capacity management rather than simply adding resources everywhere.

1. Maintain three doctors at the design load of 15 patients/hour; this is the smallest tested capacity that meets the 15-minute average-wait design target in the finite simulation.
2. Do not prioritise additional registration desks at the current workload because registration contributes only 0.42 minutes, about 6% of total waiting.
3. Treat approximately 17 patients/hour as a warning level for three doctors because utilisation becomes high and waiting rises rapidly.
4. If demand is expected to increase, consider an additional/on-call doctor during high-load periods instead of permanently overstaffing all sessions.
5. Reduce unnecessary variation in consultation duration through standardised workflows where clinically appropriate.

6. Use percentile-based waiting targets because the distribution contains a substantial long-wait tail.

## 6. Limitations and Future Work

The results should be used for comparative performance analysis rather than exact prediction of a real hospital. The study uses one 500-patient replication per scenario, so it does not provide confidence intervals. The empty-start condition and absence of a warm-up period can underestimate steady-state waiting. Configurations with $\rho \geq 1$ have no steady state and their finite-run averages should not be interpreted as stable performance.

Other limitations are the exponential service assumption, stationary arrivals, idealised continuously available servers, and exclusion of patient behaviours such as balking or reneging. The network also ends at doctor consultation and does not include laboratory, pharmacy or referral stages.

Future work should first use a warm-up period and multiple independent replications with confidence intervals. Further extensions should fit real service-time distributions, model time-varying arrival rates, detect unstable configurations automatically, and calibrate the model using real OPD timestamp data.

## 7. Conclusion

This project modelled an OPD as a two-stage tandem queue with two registration desks followed by three doctors. A Python discrete-event simulation generated patient-level data and tested changes in workload and doctor capacity. The base case showed that the doctor stage is the dominant bottleneck: 6.52 of the 6.95 average waiting minutes occurred before consultation, approximately 94% of total waiting.

The workload experiment demonstrated the nonlinear nature of congestion. Increasing demand from 15 to 20 patients/hour caused a much larger increase in waiting than the increase in arrivals itself, while the 20-patient/hour configuration was theoretically unstable. The capacity experiment showed diminishing returns after the third doctor. Therefore, improvement should focus on doctor capacity and workload control rather than additional registration capacity.

Because the dataset is synthetic and the study uses finite single runs with simplifying assumptions, the findings are best treated as evidence for comparative decisions. Warm-up periods, repeated replications and real OPD data are the most important next steps for stronger conclusions.

## References


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## Appendix

**Git repository URL: https://github.com/thamara2X1/Outpatient-Department-OPD-Healthcare-Delivery-Process**